

The IEEE TBME manuscript builds upon our earlier conference paper that reported sgACC-targeted tFUS effects on resting-state fMRI functional connectivity by substantially expanding the analysis framework and the scope of results. Whereas the conference paper primarily used paired t-tests on baseline-referenced connectivity changes and descriptive dynamic-FC trends, the TBME version introduces a prespecified subject-level linear mixed-effects difference-in-differences model to more rigorously test time-by-condition effects, explicitly evaluates and rules out baseline-driven regression-to-the-mean explanations, and adds network-level specificity analyses showing how stimulation redistributes sgACC coupling across canonical systems (notably increasing coupling with task-positive networks such as Cognitive Control rather than simply reporting global strengthening). In addition, the TBME manuscript includes new secondary analyses (e.g., testing scalp-to-target distance as a moderator), clarifies preprocessing and atlas-based time-series extraction choices, and provides expanded figures, statistical reporting, and interpretation appropriate for a full-length journal article.